CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M Sc (Biotechnology/Microbiology/Biochemistry) Examination, March 2018

Course code & Name: MS802: Molecular Biology-II

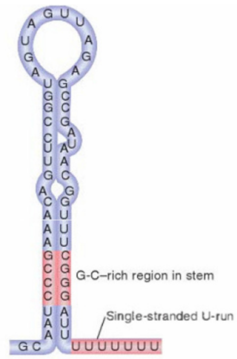
Date: 27.03.2018 Day: Tuesday Time: 01.30 pm To 03.30 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	10
1.	The consensus TATAAT denotes which of the following element in a bacterial promoter? (i) UP element (ii) -10 element (iii) -35 element (iv) none of the above	
2.	In eukaryotes which of the following RNA polymerase transcribes mRNA? (i) RNA polymerase I (ii) RNA polymerase II (iii) RNA polymerase III (iv) all of the above RNA polymerases	

3.	The figure given below best depicts the features of:  (i) rut sites (ii) nut sites (iii) intrinsic terminator (iv) none of the above	
4.	The process by which the introns are removed from the pre-RNA is called as (i) RNA splicing (ii) RNA editing (iii) RNA slicing (iv) none of the above	
5.	Which of the following factor is required for the initiation step during translation in prokaryotes? (i) IF-3 (ii) EF-G (iii) RF-3 (iv) RRF	
6.	Binding of eukaryotic mRNA to the ribosomes is facilitated by- (i) Shine-Dalgarno sequence (ii) 7-methyl guanosine cap (iii) tRNA (iv) poly A tail	
7.	siRNAs inhibit expression of a homologous gene through which of the following mechanisms? (i) Triggering destruction of its mRNA (ii) Inhibiting translation of its mRNA (iii) Inducing chromatin modifications within the promoter that silences the gene (iv) all of the above	
8.	In synthetic biology, the non-reducible elements of genetic composition (e.g. promoters, ribosome binding sites and open reading frames) are called as- (i) systems (ii) devices (iii) circuits (iv) parts	
9.	Which of the following gene encodes phage λ repressor? (i) cI (ii) cro (iii) cII (iv) all of the above	
10.	Antitermination as a means of gene regulation is observed in which of the following operon? (i) *lac* (ii) *ara* (iii) *mer* (iv) *trp*	

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Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q - II Answer the following questions as directed:		**10**
Q-II(A)	Answer **any one** of the following: (i) Write a note on: DNA Foot printing technique. (ii) What is lariat? Explain the role of lariat in pre-RNA splicing.	**04**
Q-II (B)	Answer **any two** of the following: (i) Describe the importance of -10 and -35 regions in bacterial promoters. (ii) Explain the mechanism of rho dependent termination of transcription in prokaryotes. (iii) Briefly discuss the initiation of transcription in genes transcribed by RNA polymerase II.	**06**
SECTION - II		
Q-III Answer the following questions as directed:		**15**
Q-III (A)	Answer **any one** of the following: (i) Discuss the role of aminoacyl tRNA synthetases in protein synthesis. (ii) Explain briefly the degeneracy in codons and wobble in tRNA	**04**
Q-III (B)	Answer **any one** of the following: (i) Discuss how *lac* repressor and CAP control the expression of *lac* operon. (ii) What is RNA interference? How cell uses siRNAs for gene regulation? (iii) Discuss how phage λ establishes and maintains lysogenic state.	**05**
Q-III (C)	Answer **any two** of the following: (i) Write a note on: Use of ligase independent cloning (SLIC) in synthetic biology. (ii) Giving example explain “imprinting”. (iii) Giving suitable diagram describe the control of *ara* operon	**06**